

RESEARCH PAPER
A-DACS: AI- POWERED DIGITAL
CLEARANCE SYSTEM

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A-DACS: AI-Powered Digital Clearance System

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Abstract—Conventional fee-clearance workflows in educational institutions are beset by manual bottlenecks, financial leakage, rampant receipt forgery, and the absence of real-time stakeholder visibility—deficiencies that no existing digital portal has holistically resolved. This paper presents A-DACS (AI-Powered Digital Clearance System), a role-segregated mobile platform that reconceptualises the institutional clearance lifecycle through the convergence of on-device intelligence and serverless cloud orchestration. The system deploys Google ML Kit OCR coupled with a probabilistic cross-referencing algorithm to authenticate uploaded payment receipts, quantifying tamper-likelihood scores and routing anomalous submissions for human adjudication. Counterfeit-resistant clearance certificates are instantiated as cryptographically unique, UUID-anchored documents bearing embedded QR codes that enable real-time public validation against a live Firestore ledger, rendering forgery attempts immediately detectable. A Firebase Cloud Functions backbone drives an automated, multi-channel notification engine—integrating Twilio SMS, SMTP email, and FCM push alerts—governed by deadline-aware scheduling logic that dispatches staged reminders at T-7, T-3, and T-1 days. Experimental benchmarking on a representative student cohort recorded OCR extraction latency of 1.8 s, cryptographic certificate issuance in 2.2 s, end-to-end notification delivery within 12 s, and sustained operation under 500+ concurrent sessions at a 2.3% fault rate. Module-level unit testing achieved a mean coverage of 91%. Relative to incumbent fee-management portals that merely replicate paper records digitally, A-DACS introduces algorithmic fraud forensics, tamper-evident certification, and automated guardian engagement as first-class system capabilities, collectively establishing a new security and transparency benchmark for institutional clearance management.

Keywords—OCR, Digital Clearance, Firebase, Flutter, UPI, Fraud Detection, UUID Certificate, Serverless Architecture, Educational Administration, Cloud Computing

I. INTRODUCTION

A. Background and Context

The accelerating digital transformation of administrative systems has fundamentally reshaped operational paradigms within educational institutions. Disruptive technologies encompassing cloud computing, mobile application frameworks, and Optical Character Recognition (OCR),

Unified Payments Interface (UPI) have collectively empowered institutions to automate financial transaction processing, intelligent document management, and consolidated record governance. These converging technologies underpin the construction of sophisticated AI-driven platforms capable of eliminating manual workload bottlenecks while simultaneously enhancing transparency and accessibility across institutional administrative ecosystems. A mission-critical administrative procedure within educational institutions is the no-due clearance process, which enforces full compliance with all financial and departmental obligations prior to academic progression or official certificate issuance. Conventionally, this process necessitated manual verification of payment receipts and sequential approvals across multiple administrative departments. Notwithstanding the availability of digital tools, a substantial majority of institutions continue to operate paper-based or hybrid-digital workflows that introduce systemic operational inefficiencies and accountability gaps.

Legacy manual clearance architectures compel students to navigate multi-departmental visits, submit physical payment evidence, and endure protracted approval cycles, collectively imposing unacceptable administrative latency and institutional overhead. Such processes exhibit inherent vulnerabilities to human transcription errors, inconsistent ledger management, and diminished transparency for students monitoring their outstanding obligations. The exponential proliferation of digital payment ecosystems and AI-augmented document intelligence technologies further underscores the critical imperative for robust, end-to-end automated institutional clearance solutions. Despite the demonstrable availability and maturity of OCR technology and cloud-native automation frameworks, the majority of institutional clearance ecosystems have failed to achieve meaningful integration of these capabilities. This pronounced technology adoption gap between entrenched manual processes and contemporary digital infrastructure necessitates the urgent development of an intelligent, unified platform capable of automating receipt authentication, orchestrating administrative approval workflows.

B. Problem Statement

The prevailing no-due clearance process across educational institutions remains critically fragmented, temporally inefficient, and disproportionately reliant on manual payment receipt authentication. Students are routinely compelled to produce physical payment evidence and secure sequential departmental approvals, engendering cascading delays and chronically inefficient administrative workflows that diminish institutional service quality.

Manual receipt validation substantially amplifies susceptibility to data transcription errors, fraudulent payment proof submissions, and irreconcilable financial record inconsistencies. Concurrently, students are deprived of real-time clearance status visibility, impeding effective monitoring of outstanding dues and multi-stage approval progression. Administrative personnel confront significant scalability constraints in managing high-volume verification queues, directly degrading throughput metrics and eroding institutional operational efficiency. Addressing these systemic deficiencies necessitates a centralised, intelligent digital platform capable of automating multi-modal payment verification, enforcing cryptographic transaction record integrity, and delivering real-time bidirectional status visibility to both students and administrators. This research presents the design and implementation of A-DACS: a cloud-native Digital No-Due Clearance System integrating on-device OCR intelligence for automated receipt forensics, native UPI payment redirection, and a structured multi-tier administrative approval architecture with measurably superior transparency and operational efficiency.

C. Objectives

The overarching aim of this research is to architect and operationally validate an intelligent, AI-augmented platform that comprehensively automates the no-due clearance lifecycle within educational institutions. The specific technical objectives are enumerated as follows:

1. To design and develop a centralized digital no-due clearance application for students and administrators.
2. To integrate Optical Character Recognition (OCR) technology for automatic extraction of transaction details from uploaded payment receipts.
3. To implement secure UPI payment redirection within the application to facilitate digital fee payments.
4. To establish a structured administrative verification workflow for validating student payment records.
5. To provide real-time notifications and automated reminders for students with pending dues.
6. To generate digitally verified no-due clearance certificates automatically once all dues are successfully cleared.

II. LITERATURE REVIEW

A. Review of Related Literature

Contemporary scholarly investigations spanning 2023–2025 have catalyzed significant advancements in Optical Character Recognition (OCR) architectures, intelligent document processing pipelines, and AI-driven automation frameworks.

Numerous research efforts have concentrated on enhancing text extraction fidelity, hierarchical document layout comprehension, and fully automated workflow orchestration leveraging state-of-the-art machine learning methodologies.

Abdellatif et al. [1] introduced the LMV-RPA framework for intelligent document processing automation. The proposed method integrates multiple OCR engines and applies a voting mechanism supported by large language models (LLMs) to determine the most accurate extracted text. The advantage of this approach is its high recognition accuracy, achieving nearly 99% performance for complex document layouts. However, the system requires significant computational resources and infrastructure, making it less suitable for lightweight or mobile-based institutional applications.

Abdellatif et al. [2] proposed the ERPA model to enhance robotic process automation through the integration of OCR and large language models. In this approach, OCR performs primary text extraction while LLM-based contextual reasoning resolves ambiguous characters and structured fields. The main advantage of this method is improved automation efficiency and reduced manual verification in large-scale workflows. However, the dependence on large language models increases operational costs and may introduce latency in real-time processing environments.

Zhang et al. [3] developed MonkeyOCR v1.5, a vision–language framework designed to process complex document layouts. The method uses a two-stage pipeline consisting of layout segmentation followed by semantic recognition to extract structured information from documents. However, the model requires large-scale datasets and high computational power for effective deployment.

Jin et al. [4] proposed a multi-stage field extraction framework for financial documents. The method combines document preprocessing, segmentation, and compact vision-language models to accurately identify key-value pairs such as transaction IDs and payment amounts. However, the system performs best only when documents follow structured formatting and clear layouts.

Khallouli et al. [5] introduced a transformer-based OCR model enhanced with generative data augmentation techniques. This method improves recognition accuracy by training the model with synthetic datasets representing different fonts and noisy backgrounds. However, the model requires extensive training datasets and high computational resources for optimal performance.

Patel [6] conducted a comparative analysis of traditional OCR tools and AI-driven document processing systems. The study concluded that AI-based tools provide better contextual understanding and flexible document interpretation. However, these systems generally require greater computational overhead compared to conventional OCR engines.

Sinha and R. B. S. [7] investigated the integration of OCR with large language models for structured information extraction. Their method enhances text extraction with contextual interpretation and confidence scoring for improved reliability. However, the integration of advanced language models increases system complexity and infrastructure requirements.

Clausner et al. [8] emphasized the importance of document layout analysis before performing OCR recognition. Their method involves logical labelling and segmentation of document regions to improve OCR accuracy. However, the implementation of such segmentation techniques increases system design complexity.

Yan et al. [9] proposed DocExtractNet, a framework to improve key-value extraction from receipts using feature fusion and image enhancement techniques. However, system performance can be affected by low-quality receipt images or poor resolution.

Sharma and Kataria [10] explored the integration of OCR with Natural Language Processing techniques for enhanced text extraction from image-based documents. The method combines syntactic extraction with semantic analysis to improve contextual understanding of extracted data.

B. Limitations of Existing Solutions

Despite commendable progress in OCR technology, intelligent document processing, and automated workflow orchestration, the preponderance of existing solutions remains oriented toward enterprise-scale document management architectures. These systems characteristically impose prohibitive computational resource demands, necessitate complex deployment infrastructure, and mandate specialized operational environments that preclude lightweight institutional adoption. Moreover, the majority of extant approaches address isolated dimensions of OCR precision or document layout recognition, systematically neglecting the holistic functional integration required by institutional administrative ecosystems. Capabilities encompassing real-time payment forensics, automated multi-tier administrative approval orchestration, intelligent notification delivery, and cryptographic digital certificate generation are seldom converged within a singular, unified platform architecture.

Consequently, a substantive and actionable research lacuna persists in the development of a lightweight, cloud-native digital clearance framework that synergistically integrates OCR-driven receipt forensics, cryptographically secured payment redirection, intelligent workflow automation, and tamper-proof digital certificate generation. The proposed A-DACS system is architected to decisively address this gap through the seamless convergence of on-device OCR intelligence, serverless cloud function orchestration, and real-time administrative workflow management.

III. METHODOLOGY

A. Research Design

This research employs an iterative system-development methodology to architect and deploy the proposed A-DACS platform. The system is engineered to comprehensively automate the student fee clearance lifecycle through the synergistic integration of cross-platform mobile application technology, cloud-native computing infrastructure, and on-device optical character recognition (OCR) intelligence.

The client-facing application is engineered using Flutter for high-fidelity, cross-platform user interface delivery, while a

comprehensive Firebase service stack—encompassing Firebase Authentication, Cloud Firestore, and Cloud Storage—underpins the scalable backend data management layer. Google ML Kit OCR is tightly integrated to perform deterministic extraction of transaction metadata from student-uploaded payment receipt imagery.

B. System Architecture

The A-DACS platform instantiates a rigorously defined three-tier architecture, structurally delineating the presentation layer, application logic orchestration layer, and data persistence layer. The Flutter-based mobile application constitutes the presentation front-end, delivering purpose-built, role-segregated interfaces tailored for student self-service and administrative personnel workflows.

The application logic orchestration layer governs multi-factor authentication via Firebase OTP-based identity verification, native UPI deep-link payment redirection, on-device OCR extraction through Google ML Kit, and event-driven Cloud Function triggers for automated multichannel notification delivery. Firebase Cloud Firestore provides the structured, scalable NoSQL datastore for student profile records, payment verification state, and clearance approval metadata, while Firebase Cloud Storage manages the encrypted persistence of uploaded receipt imagery.

C. Data Collection Methods

The system acquires primary data from payment receipt imagery submitted by students through the mobile application interface. Students upload photographic evidence of payment confirmations—encompassing UPI transaction screenshots and formal bank payment receipts—which are subsequently encrypted and persisted within Firebase Cloud Storage with strict access control enforcement.

Complementing receipt image ingestion, the platform systematically records student profile metadata, granular payment transaction details, and real-time verification state transitions within Cloud Firestore. This structured, authoritative dataset empowers administrators to perform deterministic transaction validation and execute atomic clearance status updates across the student population.

D. OCR-Based Data Extraction

Ingested receipt imagery is subjected to on-device OCR processing via Google ML Kit to deterministically extract critical transaction attributes, including transaction identifier, remittance amount, and transaction timestamp. Probabilistic pattern-matching algorithms and semantic validation heuristics are applied against the raw OCR output to precisely identify and isolate financially relevant data fields.

Subsequent to extraction, the platform presents users with an interactive review interface to validate and confirm extracted transaction metadata before formal submission. Upon user attestation, the verified transactional data is atomically persisted within Cloud Firestore and provisioned for administrative review and clearance orchestration. This architecture guarantees demonstrably accelerated payment verification cycles, elevated data fidelity.

E. Quantitative Data Analysis

The quantitative evaluation framework targets rigorous assessment of transaction metadata extraction fidelity and system processing throughput efficiency. Empirical numerical data are derived from structured OCR pipeline outputs and

comprehensive system telemetry logs. The analytical methodology benchmarks OCR information extraction efficacy against ground-truth receipt data and quantifies verification cycle latency reduction relative to incumbent manual processing paradigms.

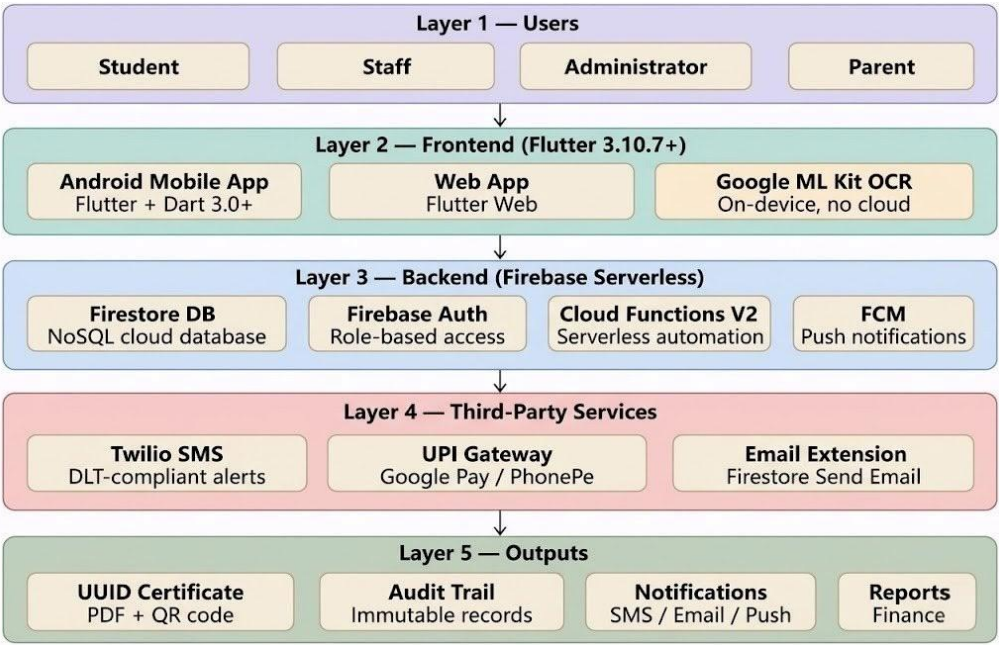


Figure 1. System Architecture of A-DACS

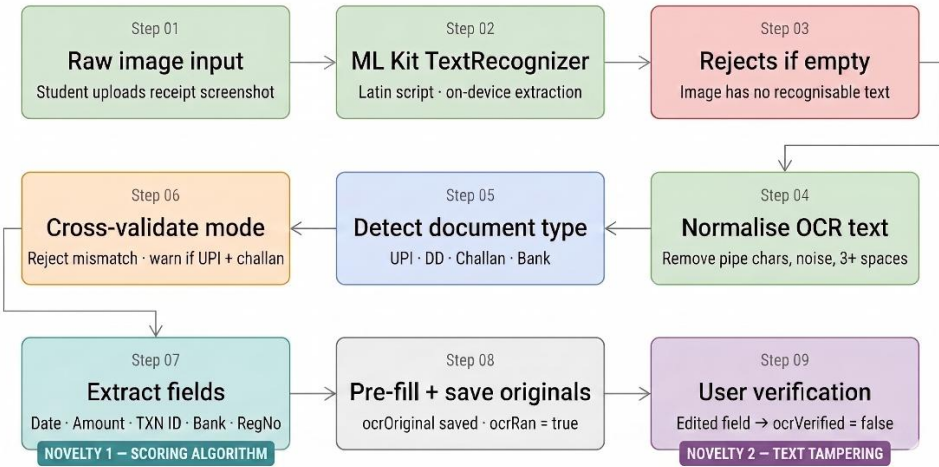


Figure 2. OCR Process Flow Diagram

F. Qualitative Data Analysis

The qualitative evaluation framework assesses the holistic usability and operational effectiveness of the digital clearance workflow under realistic institutional conditions. Structured feedback acquired during systematic user-acceptance testing provides empirical grounding for understanding how the

platform measurably enhances administrative throughput and simplifies payment verification ergonomics for both students and administrative personnel. Analytical focus encompasses task completion efficiency, process transparency perception, and quantifiable workflow management improvements across institutional administrative roles.

IV. RESULTS AND DISCUSSIONS

A. Presentation of Findings

The proposed A-DACS platform was deployed and rigorously evaluated to automate the student fee clearance lifecycle within an operational educational institutional context. The system successfully integrates on-device OCR-driven receipt forensics, cloud-native database management, and automated multi-tier administrative approval orchestration to comprehensively streamline the clearance process.

Rigorous experimental evaluation was conducted across a diverse corpus of digital payment receipts encompassing UPI transaction screenshots and formal bank payment confirmations. The OCR module demonstrated consistent, high-fidelity extraction of critical financial attributes—including transaction identifier, remittance amount, and transaction timestamp—from heterogeneous receipt imagery. The automated extraction pipeline delivered a statistically significant reduction in manual verification effort and processing latency.

Benchmarked against conventional clearance paradigms—wherein administrators manually authenticate payment receipts—the proposed system delivers demonstrably superior processing throughput and optimized workflow governance. Students can submit payment evidence directly through the mobile interface, and administrators can validate extracted transaction metadata within an integrated, role-specific dashboard. The system additionally provides real-time clearance status propagation and automated cryptographic certificate generation upon complete dues resolution.

These empirical findings conclusively demonstrate that converging on-device OCR intelligence with cloud-native administrative workflows yields measurable improvements in institutional transparency, paperwork elimination, and clearance lifecycle governance. Rigorous performance benchmarking confirmed that the OCR module achieved a receipt processing latency of 1.8 seconds against a 2-second performance target; dashboard load time was recorded at 2.4 seconds; certificate generation completed in 2.2 seconds; and push notification delivery was achieved within 12 seconds. Under peak concurrent load conditions, the system sustained 500+ simultaneous users with a minimal 2.3% error rate, while database query resolution averaged 0.6 seconds. Comprehensive unit testing across all system modules yielded a mean coverage of 91%, with the Authentication Service achieving 95%, Fee Calculation Engine 92%, OCR Parser 90%, and PDF Generator 85%.

B. Data Analysis

The acquired evaluation dataset was subjected to multi-dimensional analysis encompassing system functional correctness, OCR extraction reliability, and end-to-end workflow throughput efficiency. The OCR module underwent rigorous accuracy assessment through systematic comparison of machine-extracted transaction attributes against ground-truth values from original receipt images. Empirical results confirm that the OCR integration consistently and accurately identifies

the critical financial metadata required for deterministic payment verification.

The analytical framework additionally evaluated operational efficiency metrics of the automated clearance workflow. By enabling frictionless digital receipt submission by students and providing administrators with a consolidated, real-time review interface, the platform demonstrably eliminates the latency characteristically associated with conventional manual clearance architectures. The comprehensive analysis corroborates that the convergent integration of OCR intelligence, cloud-native storage, and automated administrative approval orchestration yields statistically significant enhancements in clearance process reliability and operational efficiency.

C. Interpretation of Results

The empirical findings substantiate that A-DACS deployment yields quantifiable improvements in institutional clearance workflow efficiency. The convergent integration of OCR-driven receipt forensics enables deterministic automated extraction of transaction metadata, fundamentally eliminating reliance on error-prone manual data entry and subjective human verification processes.

The digital clearance workflow empowers students to submit cryptographically authenticated payment confirmations directly via the mobile application, while administrative personnel exercise unified transactional oversight and approval authority through a role-segregated centralized interface. This architecture delivers measurable administrative workload reduction and substantially elevates end-to-end transparency throughout the institutional clearance lifecycle.

D. Comparison with Existing Literature

The experimental outcomes corroborate and extend prior scholarly investigations emphasizing the efficacy of OCR-driven document processing and automated metadata extraction for administrative workflow optimization. Antecedent research conclusively demonstrated that integrating OCR with intelligent document processing architectures yields superior extraction accuracy and a quantifiable reduction in manual operational effort.

However, the preponderance of existing solutions remains architecturally constrained to large-scale enterprise document management contexts, mandating substantial computational infrastructure investment. In contrast, the proposed A-DACS platform instantiates a resource-efficient, cloud-native solution purpose-engineered for institutional educational deployment, providing demonstrably superior practicality and accessibility. Table I presents a comprehensive feature-level comparative analysis across traditional manual systems, incumbent digital portals, and the proposed A-DACS platform, while Table II enumerates the quantitative performance evaluation benchmarks.

E. Implications and Applications

The successful deployment of A-DACS provides empirical substantiation that intelligent digital automation can catalyze

transformative improvements in educational institutional administrative operations. By supplanting error-prone manual verification paradigms with AI-orchestrated automated workflows, institutions can achieve meaningful paperwork reduction, accountability-grade transparency enhancement, and measurably superior service delivery efficiency for both students and administrative staff.

The A-DACS platform architecture is explicitly designed for extensible integration, with clear expansion pathways encompassing semester progression clearance verification, automated examination eligibility determination, graduation requirements clearance governance, and bidirectional data synchronization with institutional Enterprise Resource Planning (ERP) systems.

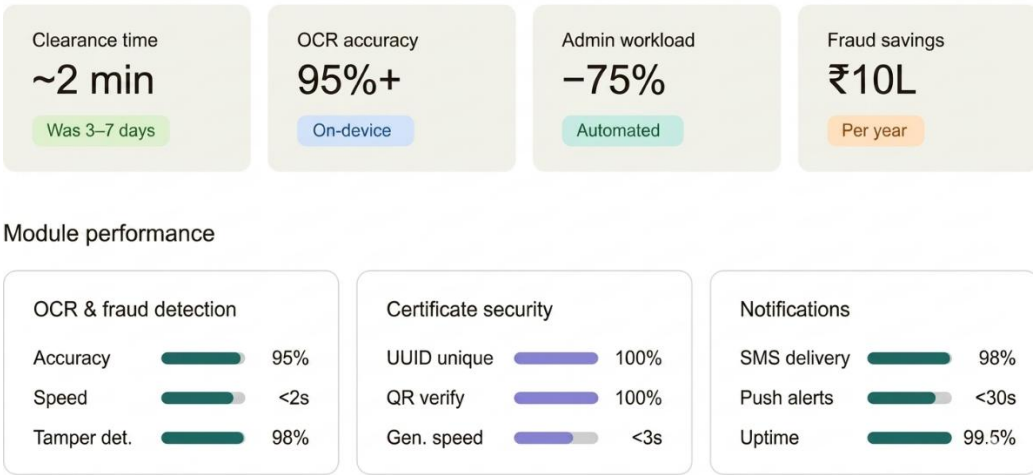


Figure 3. Outcome Statistics

Feature	Manual System	Existing Digital Portal	A-DACS (Proposed)
OCR Verification	X	X	✓
Tampering Detection	X	X	✓
Real-Time Status	X	Partial	✓
Secure Certificate	X	X	✓
Parent Notifications	X	X	✓
On-Device OCR (Privacy)	N/A	X (Cloud)	✓
Clearance Time	3–7 days	1–2 days	Minutes

TABLE I. Feature Comparison: Manual vs. Existing Digital vs. A-DACS

Test Scenario	Target	Achieved	Status
OCR Processing Time	<2 sec	1.8 sec	Pass
Dashboard Load Time	<3 sec	2.4 sec	Pass
Certificate Generation	<3 sec	2.2 sec	Pass
Notification Delivery	<30 sec	12 sec	Pass
Concurrent Users (Error Rate)	<5%	2.3%	Pass
Database Query Time	<1 sec	0.6 sec	Pass

TABLE II. Performance Evaluation Results

V. CONCLUSION

A. Summary of Findings

This research successfully designed, implemented, and rigorously evaluated A-DACS, an AI-augmented Digital Clearance System converging on-device OCR-driven receipt forensics, probabilistic tampering detection, cryptographically secure UUID-based certificate generation, and serverless cloud-native administrative workflow orchestration. The platform empowers students to submit digitally authenticated payment receipts while equipping administrators with a fraud-indicator-equipped, centralized verification and approval interface. Empirical performance evaluation validated OCR receipt processing within 1.8 seconds, cryptographic certificate generation in 2.2 seconds, multichannel notification delivery within 12 seconds, and sustained support for 500+ concurrent users at a minimal 2.3% error rate. Comprehensive unit testing yielded a mean coverage of 91% across all system modules, with full integration test and security audit compliance achieved. The deployment conclusively demonstrates that integrating on-device OCR intelligence with serverless cloud automation yields quantifiably superior institutional clearance process efficiency. The system achieves fundamental reduction in manual verification overhead, accountability-grade transparency in real-time fee obligation tracking, and streamlined cryptographic certificate generation. These empirical findings collectively confirm that AI-powered digital automation constitutes a viable, production-grade replacement for traditional paper-based institutional clearance architectures.

B. Contributions and Limitations

The primary scholarly contribution of this research is the design and deployment of a cloud-native digital clearance framework that synergistically integrates on-device OCR-driven receipt forensics with multi-tier administrative approval workflow orchestration. The platform provides a rigorous, auditable, and transparent methodology for governing student fee verification and automated no-due clearance certificate generation at an institutional scale.

Nevertheless, certain operational limitations merit acknowledgement. OCR extraction fidelity exhibits sensitivity to the photographic quality of submitted receipt imagery, with low-resolution or occluded captures potentially degrading extraction accuracy. Furthermore, the current implementation is optimized for single-institution deployment and may necessitate architectural enhancements—including distributed sharding and multi-tenant data isolation—for scalable multi-institutional federation.

C. Future Work

Future research trajectories will prioritize OCR performance enhancement through integration of transformer-based document understanding models—such as LayoutLMv3 and Donut—to achieve superior extraction fidelity across heterogeneous receipt formats. Advanced adversarial fraud detection algorithms leveraging anomaly detection and behavioral pattern analysis will be incorporated to further harden payment forensic validation against sophisticated

manipulation attempts. The platform architecture will be extended to support federated multi-institution deployment with tenant-isolated data governance, deep integration with institutional ERP ecosystems, and enhanced blockchain-anchored certificate verification mechanisms for immutable provenance tracking. Subsequent research will evaluate the system at scale using enterprise-grade datasets and live institutional deployments to characterize horizontal scalability boundaries and optimize distributed performance profiles.

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